Mohammad Najeeb

Frankfurt, Germany | +49 1573-0050-253 | mona00002@uni-saarland.de
github.com/mohammadnajeeb | linkedin.com/in/mohammadnajeeb

EDUCATION

M.Sc. Computer Science (Specialization: Media) — Apr. 2025 – Apr. 2027
Universität des Saarlandes — *Saarbrücken, Germany*

B.Sc. (Hons.) Mathematics — Aug. 2019 – Jul. 2022
Aligarh Muslim University — *Aligarh, India*

EXPERIENCE

Research Assistant (HiWi) — Apr. 2026 – Present
Educational Technology Department, Saarland University — *Saarbrücken, Germany*

SRL Course Platform (Full-Stack, Lead Developer)

- Architected end-to-end adaptive learning platform (React + TypeScript, Node.js/Express, PostgreSQL + pgvector); authored ADRs covering authentication, LLM abstraction, data architecture.
- Engineered a Medallion ETL pipeline ingesting PDFs, LLM-driven semantic chunking, and pgvector HNSW indexing.
- Implemented hybrid retrieval combining pgvector HNSW vector similarity and BM25 full-text search (PostgreSQL GIN index) via Reciprocal Rank Fusion (RRF), exposed through a FastAPI endpoint.

AI Co-Design Platform — Audio-to-Knowledge-Graph Pipeline

- Architected NLP pipeline converting pedagogical audio to knowledge graphs: OpenAI Whisper STT, pyannote speaker diarization, LLM entity and relation extraction.

Real-Time Gaze Streaming Optimization

- Optimized WebSocket based real-time gaze data streaming from Pupil Labs glasses API, reducing end-to-end latency on a custom behavioral data collection tool.

Research Assistant (HiWi) — Sept. 2024 – Mar. 2025
Computational Interaction Group (CIX), Saarland University — *Saarbrücken, Germany*

- Built automated computer vision pipeline (OpenCV, MediaPipe) processing 50–60 typing videos (5M+ frames); achieved 96% accuracy on hand posture detection, eliminating manual annotation bottleneck for interaction research.
- Delivered production-ready dataset and reusable pipeline enabling scalable behavioral analysis for touchscreen/gesture interaction studies.

Web Developer & UI/UX Intern — Jul. 2022 – Sept. 2023
Orion Solutions — *Lucknow, India · Hybrid*

- Shipped responsive web interfaces (React, HTML, JavaScript) in cross-functional team; managed design-to-dev handoff; integrated REST APIs and database layers (MongoDB, SQL).

TECHNICAL SKILLS

- **Languages:** Python, TypeScript, JavaScript, SQL
- **LLM & Generative AI:** RAG pipelines, LLM fine-tuning, prompt engineering, vector databases, agentic workflows, LangChain,
- **ML Frameworks:** PyTorch, TensorFlow, scikit-learn
- **Backend & APIs:** Node.js/Express, FastAPI, REST API
- **Databases:** PostgreSQL, pgvector, MongoDB
- **DevOps & MLOps:** Docker, Git, Linux, CI/CD

ADDITIONAL PROJECTS

- **NCBI Gene MCP Client (Python, FastAPI, MCP)** — Built production-ready MCP server integrating NCBI Entrez API; JSON-RPC compliant, deployed on Vercel.
- **UniChatbot (LangChain, ChromaDB, FastAPI, Docker)** — Engineered RAG chatbot for 500+ university queries; 35% improvement in query relevance via semantic search.

PUBLICATIONS

Zea Mays Leaf Disease Classification Using Swin Transformer — *Springer (SIGMAA 2024)* [DOI: 10.1007/978-981-97-6352-8_59]